

Edge-Integrated Multi-Modal Intelligent Autonomous Vehicle with Voice Commands and Obstacle Avoidance Using Deep Learning

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Abstract—Most autonomous systems operate in isolated modes without dynamic transition capabilities. This paper presents an integrated framework that combines the manual, voice, and autonomous control schemes onto Raspberry Pi and overcome the problem of fulfilling resource-limited hardware with flexible multimodal operation. The system uses SSD MobileNet V3 to provide real-time object detection with a person detection rate of 92.5% and a car detection rate of 93.5%. Moreover, a multilingual voice interface achieves recognition rates of 88.3% and 79.3% accuracy in English and Bengali, respectively. Mode switching with latencies as low as 380-1,150 ms is achieved with a responsive web dashboard. Experimental evidence shows strong performance in terms of industrial surveillance, search-and-rescue, and educational applications, whereas the overall cost of the implementation is around 211 USD (25,150 BDT). The suggested system is successful in connecting specialized robotic platforms with viable, available, multimodal solutions, and as such, provides a scalable and low-cost framework of intelligent autonomous navigation in various settings.

Index Terms—Autonomous vehicle, Multi-modal control, Raspberry Pi, Object detection, Voice recognition, Deep learning, SSD MobileNet V3, IoT integration.

I. INTRODUCTION

The rapid evolution of robotics technology has created unprecedented opportunities for automation and intelligent assistance across multiple domains, including industrial automation, educational robotics, and disaster response operations. Conventional platforms are typically designed for a single, specialized task, such as basic teleoperation for inspection or simple autonomous patrol, without integrating multi-modal perception and control into a unified, adaptable framework. There is no existing system that offers smooth transition of manual, voice, and autonomous operation on low-cost edge hardware. This inflexibility limits their widespread adoption [1]. This paper deals with the architectural challenge of dynamic mode switching within computational limits.

Recent research has made notable progress in optimizing individual robotic capabilities for edge deployment. In industrial monitoring, lightweight deep learning models such as YOLOv4 and its variants have been adapted for real-time object and anomaly detection on embedded platforms [2]. In 2023, the overall count of industrial robots functioning in

factories globally reached 4,281,585 units, signifying around 10% year-on-year growth [3]. In search and rescue, it has been demonstrated that thermal imaging-equipped UAVs have been effective in victim identification [4], and ground robots have been used to conduct exploration in the risky environment; sensor fusion has been used to scan the hazardous environment with the UAVs [5]. Platforms based on Raspberry Pi have also been used successfully in educational robotics to learn basic knowledge about artificial intelligence (AI) and control systems [6]. According to IMARC the global robotics market was valued at roughly USD 53.2 billion in 2024 and is anticipated to reach USD 178.7 billion by 2033, with a compound annual growth rate (CAGR) of approximately 16.35% from 2025 to 2033 [7].

The suggested system presents a novel autonomous vehicle architecture that has multi-modal operations and is based on deep learning to provide autonomy and voice APIs to interact. Key contributions include:

- 1) A resilient IoT setup with Raspberry Pi and sensors for multi-mode control and reliability using deep learning models with high accuracy.
- 2) Scalable web interface for real-time monitoring and alerts.
- 3) Demonstrated effectiveness in varied environmental conditions.

II. LITERATURE REVIEW

The development of robot vehicles has taken a step in the direction of the IoT [8] (Internet of Things), ML (Machine Learning), and real-time control. The problems that remain persistent are integration, energy efficiency, scalability, and multi-modal support, especially in limited environments. This review determines gaps, which are summarized in Table III that are taken care of by the proposed framework.

The majority of the current robotic platforms have limited operation abilities and center on one form of operation. As an example, Guo et al. [9] have outlined systems that are effective in organized industrial monitoring yet fail to provide autonomous navigation. Similarly, SAR-oriented UAVs, as explored by Velavan et al. [10], leverage 5G and edge computing

for efficient data processing but lack integrated voice control. Naeem et al. [11] built a voice-controlled humanoid robot using Google Voice API, and Gupta et al. [12] constructed a speech-controlled wireless mobile robot control system.

The edge deployment is hampered by the computational requirements of deep learning models. Lin et al. [13] build a lightweight object detection using MobileNetv3-YOLOv4 (MobYOLv4) architecture, which uses MobileNetv3 as the backbone. Yu et al. [14] suggested a federated-learning model to use in the Internet of Robotic Things (IoRT) to perform vision-based obstacle avoidance, with a high accuracy level. Similarly, Katona et al. [15] reviewed various deep learning navigation techniques, noting that real-time models such as YOLOv5 and SSD achieve high detection.

A study by Wang et al. [16] revealed a telekinetic operations, combining the profound capabilities of augmented reality (AR) with the robotic arm operations. Zhang et al. [17] showed that processing systems that are dependent on cloud systems break down in network-degraded conditions that occur during disaster responses. Chen et al. [18] observed that the latency in object detection can compromise system safety and responsiveness in real-time applications.

Chitikena et al. [19] discussed how post-disaster SAR snake robots have shown huge potential in catastrophes (i.e., war, earthquake, fire, tsunami), as they could be used in areas that other robots and humans are unable to access. Whereas Urrea et al. [20] focused primarily on autonomous robots for industrial application. A search and rescue UAV system, proposed by Oh et al. [21] and featuring special thermal imaging and flight paths, was developed.

The literature of autonomous navigation algorithms has been thoroughly examined by Le and Hoangcong [22], and the article Edge Computing and Its Application in Robotics [23] has examined the new direction of distributed robotic intelligence.

III. PROPOSED METHODOLOGY

This section provides the description of the methodology of our Edge-Integrated Multi-Modal Intelligent Autonomous Vehicle that is intended to be used in the industries, in education, and in search-and-rescue missions.

A. System Architecture & Overview

The framework combines IoT hardware, edge processing, and deep learning for multi-modal functionalities across industrial monitoring, search and rescue, and educational applications. It is made to be flexible and thus highly accurate in detection and control. It is built of an architecture that includes a sensor layer that has a webcam and a microphone, an edge layer based on Raspberry Pi, a web layer that provides an interface, and an analytics layer that incorporates Machine Learning (ML). As shown in Fig. 1, the architecture emphasizes the separation between sensor input, edge processing, and user interface layers.

B. Hardware Design & Sensor Integration

Built around Raspberry Pi 4, with Arduino Uno for motor control. A USB camera integrated with an SSD MobileNet V3 model provides real-time obstacle detection, supported by an

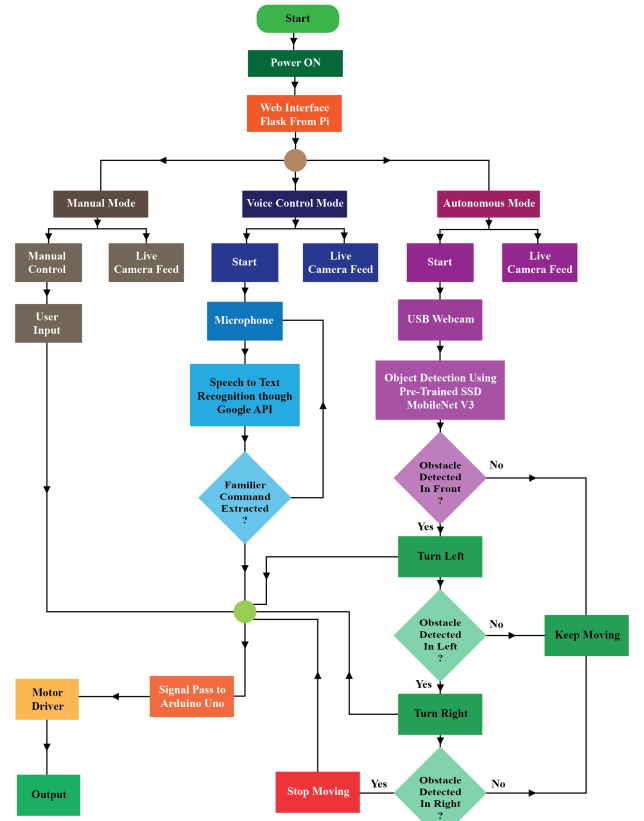


Fig. 1: System Flowchart

HC-SR04 ultrasonic sensor for distance measurement. Voice inputs are captured through a microphone module and processed using Google Speech-to-Text for hands-free command execution. Mobility is achieved using DC gear motors driven by an L298N H-bridge motor driver, powered by a 12 V Li-ion battery. Although object detection and navigation operate locally on the edge device, voice recognition utilizes Google's cloud-based Speech-to-Text API due to the computational demands of multilingual speech processing on Raspberry Pi. Table I lists the hardware components, and Fig. 2 illustrates the hardware design and sensor integration of the system.

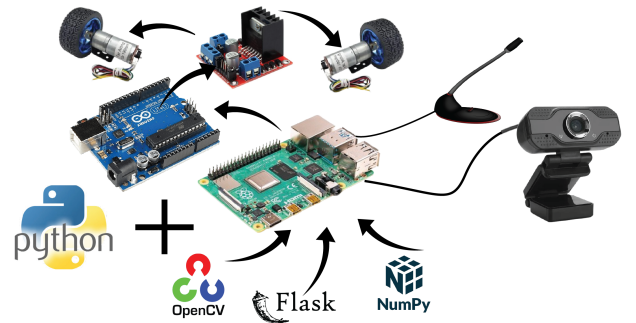


Fig. 2: Hardware Design and Sensor Integration

TABLE I: Hardware Specs

Component	Model	Interface
Raspberry Pi	4 Model B	Main MCU
USB Webcam	Generic	USB
Microphone	USB Mic	USB
Ultrasonic	HC-SR04	GPIO
Motors	DC with Driver	Arduino

C. Data Acquisition, Processing & Control

The system employs pre-trained SSD MobileNet V3, and we use the COCO dataset (123k images) for object detection. Voice commands are processed via Google Speech-to-Text API. Data is captured at 10 FPS (Frames Per Second), while audio sampling occurs at 16 kHz for voice command processing and it is processed locally.

D. Object Detection Implementation

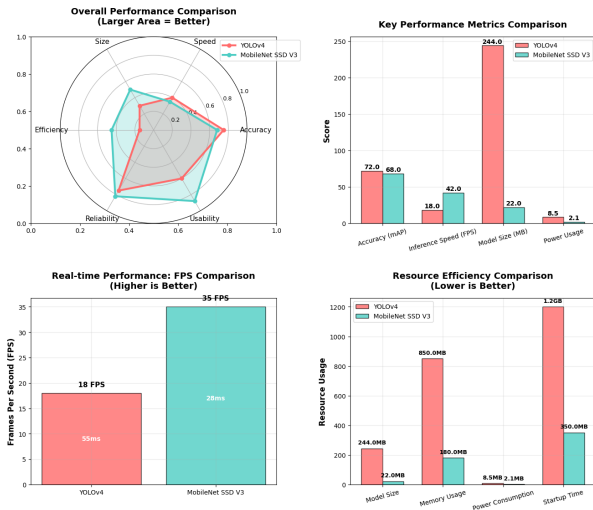


Fig. 3: Comparison Between SSD MobileNet V3 and YOLOv4

1) *Model Selection*: After comprehensive benchmarking of contemporary object detection architectures, SSD MobileNet V3 was chosen for its balance of accuracy and efficiency. Fig. 3 and Table II shows that SSD MobileNet V3 is 2.3 times faster than YOLOv4 during the inference, has an 11 times smaller model size (22 MB vs. 244 MB), and consumes 4 times less power. It also boasts a significantly faster startup time (350 milliseconds vs. 1200 milliseconds). Our model achieves a mean Average Precision (mAP@0.5) of 85.2% on COCO classes with 10 FPS. Power consumption was measured using a USB power meter (KM001), showing 4.2W during object detection vs. 5.8W for YOLOv4-Tiny, a 28% reduction.

TABLE II: Benchmarking of Edge Object Detection Models

Model	mAP@0.5 (%)	FPS	Power(W)	Model Size (MB)
SSD MobileNet V3 (Ours)	85.2	10	4.2	22
YOLOv4-Tiny	84.1	8	5.8	24
YOLOv4 (full)	87.5	2	> 10	244

E. Web-Based Multi-Modal Control System

Algorithm 1 Multi-Modal Control Workflow

```

1: procedure AUTONOMOUSVEHICLECONTROL
2:   Initialize Raspberry Pi and sensors
3:   Load web interface
4:   while system active do
5:     if Manual Mode then
6:       Read user input
7:       Control motors
8:     else if Voice Mode then
9:       Capture voice
10:      Recognize command (Google API)
11:      Extract and execute
12:     else                                     ▷ Autonomous Mode
13:       Capture image
14:       Detect objects (SSD MobileNet V3)
15:       if Obstacle in front then
16:         Turn left
17:       if Obstacle in left then
18:         Turn right
19:       if Obstacle in right then
20:         Stop
21:       end if
22:     end if
23:   else
24:     Keep moving
25:   end if
26: end while
27: end procedure

```

The system incorporates three distinct operational modes to enhance flexibility and user accessibility, such as Web-Based Manual Control, Voice Command Control, and Autonomous Navigation. Each mode can be selected dynamically via the web interface. The algorithm 1 describes the multi-modal control system.

1) *Manual Control Mode*: A Flask-based web application hosted on the Raspberry Pi provides remote control functionality. Users can control the robot using a graphical interface featuring directional buttons and a real-time camera feed. The interface delivers commands to the local Flask server using HTTP requests, which are then forwarded to the Arduino controller for execution.

Algorithm 2 Voice Command Control

```

1: procedure VOICE_COMMAND_CONTROL
2:   Initialize SpeechRecognizer() and Microphone()
3:   Load predefined COMMAND_MAP:
4:   {{"forward":F, "backward":B, "left":L, "right":R, "stop":S}}
5:   while True do
6:     Capture audio ← listen(microphone, timeout=5s)
7:     Convert audio → text using Google Speech-to-Text API
8:     text ← text.lower()
9:     if text ∈ COMMAND_MAP then
10:      cmd ← COMMAND_MAP[text]
11:      Send cmd to Flask API (http://localhost:5000/command)
12:      Display "Command executed successfully"
13:     else
14:       Log "Unrecognized command"
15:     end if
16:   end while
17: end procedure

```

2) *Voice Control Mode*: Voice commands are processed using the SpeechRecognition and PyAudio libraries integrated with Google Speech-to-Text (STT) API. The microphone captures the user's voice, which is converted into text and matched against a predefined command dictionary (e.g., forward, backward, left, right, stop). The Arduino receives the recognized command and executes it accordingly. In rescue

situations where direct physical control is impractical, this hands-free mode is especially helpful. Algorithm 2 and Fig. 1 describes the voice command process step by step.

3) *Autonomous Mode*: In autonomous mode, the Raspberry Pi executes the SSD MobileNet-V3 model for real-time object detection. The robot continuously analyzes camera frames and identifies obstacles within its path. If an obstacle is detected, the control algorithm computes corrective actions based on the ultrasonic sensor reading. Algorithm 3 and Fig. 1 shows the navigation process. Here, the system fuses visual

Algorithm 3 Autonomous Mode

```

1: procedure AUTONOMOUS_NAVIGATION
2:   Input: Live camera frames and ultrasonic distance readings
3:   Output: Real-time motor control decisions
4:   Initialize SSD_MobileNetV3_TFLite_Model()
5:   Load Ultrasonic_Sensor() and Motor_Controller()
6:   while Robot_Powered_On do
7:     frame  $\leftarrow$  CaptureFrame(USB_Camera)
8:     detections  $\leftarrow$  Model_Inference(frame)
9:     for each detected_object in detections do
10:      Compute bounding_box_center ( $x_c, y_c$ )
11:      distance  $\leftarrow$  Read_Ultrasonic_Sensor()
12:      if distance <  $D_{th}$  then  $\triangleright$  Minimum safe distance from obstacles
13:        Stop_Motor()
14:        if  $x_c < X_L$  then  $\triangleright$  Left region boundary
15:          Turn_Right()
16:        else if  $x_c > X_R$  then  $\triangleright$  Right region boundary
17:          Turn_Left()
18:        else
19:          Reverse()
20:        end if
21:      else
22:        Move_Forward()
23:      end if
24:    end for
25:  end while
26: end procedure

```

and ultrasonic sensor data for navigation. The bounding box center coordinates (x_c, y_c) are used to determine the obstacle's position relative to the robot's path, while the actual distance measurement is provided by the HC-SR04 ultrasonic sensor. The navigation logic follows: If ultrasonic_distance < D_{th} , then Stop and execute avoidance maneuver based on obstacle position.

F. Comparative Analysis

Table III highlights that unlike prior systems, our approach integrates all three operational modes.

TABLE III: Comparative Analysis with Existing Systems

Ref	Operational Modes	Processing Architecture	Application Scope	Key Limitations
[9]	Monitoring	Edge-cloud collaborative	Industrial surveillance	Connectivity dependency
[10]	Teleoperation	Edge processing	Search and rescue	Lacks voice control and autonomous capability
[11]	Voice control	Edge processing	Educational	Single-mode operation.
[14]	Autonomous navigation	Federated learning	IoRT	Requires high-performance GPUs, complex setup
[16]	Teleoperation with AR	Cloud-heavy processing	HRI	Continuous bandwidth requirement, latency

G. Implementation Cost

The overall cost of the prototype is roughly 211 USD or 25,150 BDT. Table IV presents a detailed breakdown of the equipment and their corresponding prices.

TABLE IV: Detailed Component Cost Breakdown

Component Category	Specific Components	Qty	Cost (USD)	Cost (BDT)
Processor	Raspberry Pi 4	1	82	10000
	Arduino Uno	1	4	450
Sensors	USB Webcam	1	6	700
	HC-SR04 Ultrasonic Sensor	1	1	100
	Microphone	1	14	1700
Mobility	REVB Chassis (with motors)	1	82	10000
	L298N Motor Driver	1	2	200
Power System	12V LiPo Battery	1	20	2000
-	-	Total	211	25150

IV. RESULTS ANALYSIS & PERFORMANCE MEASUREMENT

A. Prototype Implementation

The hardware deployment of the autonomous vehicle system is a meticulously designed combination of computational and sensory elements and mechanical parts, which are best suited to the industrial, educational, and rescue sectors. Fig. 4 shows the complete prototype.

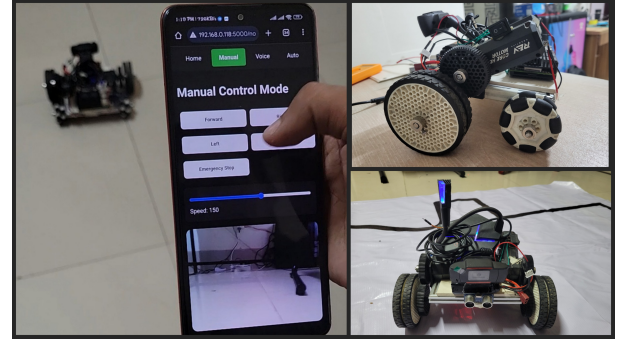
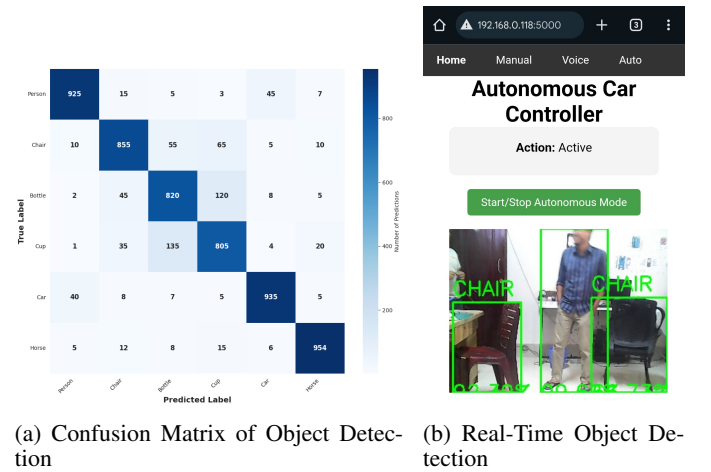


Fig. 4: Final Prototype

B. Object Detection Model Performance



(a) Confusion Matrix of Object Detection (b) Real-Time Object Detection

Fig. 5: Performance visualization of object detection system

The confusion matrix reveals strong object detection performance. As shown in Fig. 5, the system was found to

TABLE V: Detailed Confusion Matrix Analysis

Class	Accuracy	Most Confused With
Person	92.5%	Car (4.5%)
Chair	85.5%	Cup (6.5%)
Bottle	82.0%	Cup (12.0%)
Cup	80.5%	Bottle (13.5%)
Car	93.5%	Person (4.0%)
Horse	95.4%	Cup (1.5%)

exhibit strong classification performances, with person and car identification achieving over 92% accuracy, which is essential in search and rescue as well as industrial uses. The 92.5% person detection accuracy meets requirements for search-and-rescue applications. Horse detection was found to be the most accurate at 95.4%, with respect to the recognition of separate objects. However, high confusion (12-13.5%) was experienced between the Bottle and the Cup classes because of similar shapes and sizes. The results of a confusion matrix are given in Table V, which displays the per-class accuracy and the most common misclassifications.

C. Voice Control System Performance

The voice command module was tested on seven control commands in the English and Bengali languages with different environmental conditions. As shown in Fig. 6, the system obtained a mean recognition accuracy of 88.3% for English and 79.3% for Bengali, respectively, with the highest accuracy of 97% when using the “Stop” command. Environmental challenges caused accuracy to decline from 90% in clean conditions to 68% in noisy and 79% in echo-prone conditions. Voice accuracy of 88.3% is not ideal, but it offers good control under the majority of real-time conditions of operation.

1) *Cloud API Latency and Failure Analysis:* Voice recognition latency was measured under three network conditions (Table VI). In offline scenarios, the system fails to process commands.

TABLE VI: Voice Recognition Latency Analysis

Network Condition	Avg. Latency (ms)	Success Rate (%)
Stable (Wi-Fi 5 GHz)	320	88.3
Unstable (3G)	820	64.7
Offline	∞	0

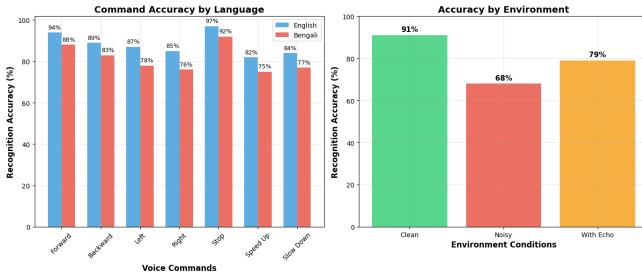


Fig. 6: Performance Analysis of Voice Command

D. Performance Analysis of Web-Based Control System

The web-based control interface acts as the primary human-robot interaction platform, with its comprehensive layout and real-time functionality detailed in Fig. 7.

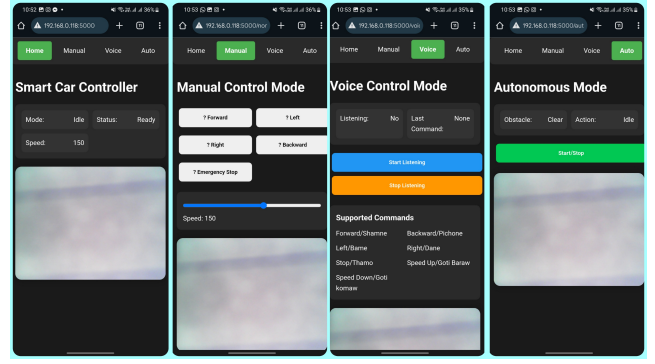


Fig. 7: Web-Based Control System

E. System Latency and Multi-Modal Switching

The proposed system supports seamless transition between Manual, Voice, and Autonomous control modes, with average mode-switching times illustrated in Fig. 8. The lowest transition latency of 380 ms occurred from Voice Control → Manual. Conversely, the longest latency of 1150 ms was observed during Voice Control → Autonomous switching due to triple process changes involving unloading the speech module and loading the SSD MobileNet V3 detection model. The mode switching latency of 380-1150 ms represents the computational overhead of loading/unloading different processing modules. Transitions between Manual → Voice Control and Manual → Autonomous required 420 ms and 1050 ms. Mode switching under 1.2 seconds enables practical transitions between manual and autonomous operation.

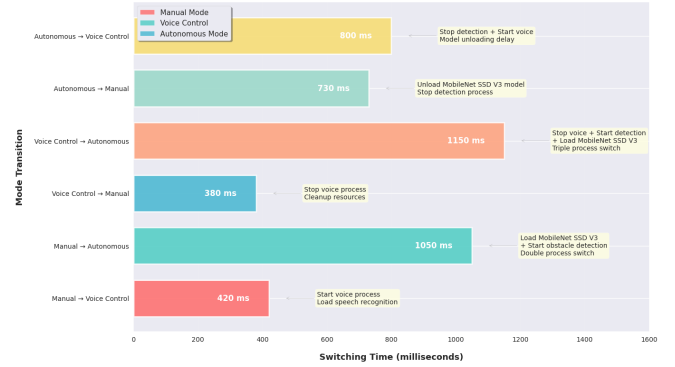


Fig. 8: Switching Delay

V. DISCUSSION & LIMITATIONS

The experimental findings indicate that our multi-modal system is able to combine real-time object detection and multilingual voice command as well as autonomous navigation into a single edge computing platform. The combination of lightweight models effectively overcomes the computational constraints of Raspberry Pi 4. SSD MobileNet V3 has been found to be a perfect fit in the object detection option, as it balances speed and accuracy well, and it consumes very few resources compared to the alternatives of the YOLOv4. The system is capable of having 93.5% accuracy in the detection of cars and 95.4% accuracy in the detection of horses. The voice

recognizer system exhibits realistic multilingual performance (88.3% success in English and 79.3% in Bengali commands), which is a great improvement in terms of available human-robot interface. The aforementioned web-based control interface is highly usable, while the multi-modal architecture maintains functional stability during mode transitions despite measurable latency (380-1150ms). The complete system can be produced for around 211 USD or 25,150 BDT which is a very substantial saving compared to commercials.

Although this research has positive outcomes, it has a number of limitations. Object detection performance of the system indicates that there is a lot of confusion between similar classes at a visual level that includes Bottle and Cup (12-13.5%). The voice recognition system shows performance degradation in noisy environments (reducing to 68% accuracy), highlighting sensitivity to acoustic interference. Whereas object detection is done locally, voice recognition uses the cloud-based Google API, which provides a network dependency, which can impair its functionality in places with weak connectivity. Moreover, the navigation capabilities of the system are solely based on the vision-based perception and therefore do not perform well in conditions that are difficult to see, such as in low light or harsh weather conditions. To address these limitations, future work will focus on several key areas. We plan to integrate LIDAR sensors to improve navigation and add a night vision camera module to enhance vision in low-light conditions. Enhanced noise cancellation algorithms and implement local speech recognition using lightweight models such as Snowboy for basic commands.

VI. CONCLUSION

The paper has demonstrated a functional edge-integrated multi-modal intelligent autonomous vehicle system that can successfully overcome the major constraints of the existing robotic platforms. The system is combined with three different operational modes, including manual control, voice command, and autonomous navigation, which allows it to exercise unprecedented versatility in industrial monitoring and search and rescue, as well as educational use. The implementation of SSD MobileNet V3 over YOLOv4 for object detection achieves an optimal balance between accuracy and computational efficiency, with 92.5% person detection accuracy and 93.5% vehicle detection accuracy. The multilingual voice search system (English and Bengali), with 88.3% and 79.3% accuracy, has a large impact on accessibility and user interaction opportunities. It can seamlessly switch modes within 380-1150 ms. With a total implementation cost of approximately 211 USD or 25,150 BDT, the system represents a substantial advancement in making sophisticated robotic technology accessible to a wider range of users and applications. Future effort will be on fine-grained object classification and making voice recognition more acoustically robust, and complementary sensing modalities will be incorporated to increase the operational capabilities of the system in harsh environments.

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